The Process: A First-Class Analytical Unit

Conditional-Information Signal Selection and the Effective Degrees of Freedom of an Alpha Book

`process_preprint.tex` · 14pp · Jul 2026 · renamed from `process_related` 2026-08-08 · methods/framework paper, empirics stated as future work

Thesis

A microstructure feed produces hundreds of features, most mutually redundant. The operative question is therefore **not** whether feature X predicts, but **how much non-redundant information the set carries in aggregate**, and **how many statistically independent bets that supports**.

The unit

Beside the **feature** (what is computed) and the **algorithm** (how it trades), place a third first-class unit: the **process** — a typed, registry-bound analytical contract that certifies whether a feature, or a derivative/combination of features, carries information about a *precisely specified* target. Every process ships behind a mandatory planted-signal conformance gate.

Why information, not correlation

Mutual information as the currency of prediction, justified by **Fano's inequality** — information is a hard *lower bound* on achievable error. A feature with zero MI about the target cannot be traded by any model, however clever; correlation gives no such guarantee.

The central quantity

Fundamental law of active management:

$$\boxed{\mathrm{IR} \approx \overline{\mathrm{IC}}\sqrt{N_{\text{eff}}}}, \qquad N_{\text{eff}} = \frac{N}{1+(N-1)\bar{\rho}}$$

The binding quantity is the stress-regime correlation, not the calm-regime one. A book of 20 signals with $\bar{\rho} = 0.1$ in calm and 0.8 in stress has $N_{\text{eff}} \approx 6.9$ when nothing is happening and $N_{\text{eff}} \approx 1.2$ exactly when it matters. Reporting calm-regime breadth is the standard way to lie to yourself about diversification.

Five orthogonalization mechanisms, each a process

1. **Conditional-MI greedy selection** — the core extractor. Add the feature maximising $\mathrm{I}(X;Y \mid S)$ given what is already selected.
2. **Interaction-information synergy mining** — finds pairs that carry jointly what neither carries alone (negative interaction information); the thing greedy marginal screens structurally cannot see.
3. **Training-prefix residualization** — pure-innovation extraction, fit on the prefix only so the residual is causal.
4. **Marchenko–Pastur-denoised PCs** — separates real factors from the eigenvalue bulk that pure noise produces at finite T/N. Tells you how many factors are *real.*
5. **Capstone: a decorrelated signal book that reports its own honest N_{eff}.**

Making "price action" explicit

An aspect × horizon target taxonomy — a process must name *which* aspect (sign, magnitude, volatility, timing) at *which* horizon it certifies. Half the ambiguity in feature research is a target that was never written down.

Minting new degrees of freedom

A derivative-operator layer creates new feature DoF from existing ones: fractional differentiation (memory-preserving stationarity), leaky integration, causal spectral features — all under a **program-level FDR ledger**, because minting features is exactly how a search silently multiplies its own multiple-testing burden.

Estimation hygiene

Keeping the breadth estimate honest is its own section: N_{eff} computed from a noisy $\hat{\rho}$ is biased upward, so the paper treats the estimate itself as something requiring a gate rather than a formula you evaluate once and quote.

Implementation status (NAT)

PROC critical path complete 2026-07-30: null-calibration, `conditional_predictability`, standing 3-bar eval, process-FDR + cross-run ledger, `horizon_label_scan`, predictability surface (`nat viz predictability`). **[!]** First real surface: 284 BTC cells, argmax $z = 13.2$ but $q = 0.56 \Rightarrow$ correctly [SPEC], **0 discoveries**. BH at the 100-shuffle p-floor over 284 cells *cannot* pass — real discoveries need deeper nulls ($n_{\text{shuffles}} \gtrsim m/\alpha$) on focused candidate sets, or more clean data. The zero is a power problem, not a verdict on the features.

The two kinds (they share registry, CLI, persistence)

Evaluation process — scores existing columns, produces no new series:

$$\mathcal{E} : (\mathbf{D}, \mathcal{C}) \longmapsto \mathcal{R}, \qquad \mathbf{D} \in \mathbb{R}^{T\times p}$$

Transform process — mints m new series on the input index (PCs, residuals, fractional differences):

$$\mathcal{T} : (\mathbf{D}, \mathcal{C}) \longmapsto (\mathbf{D}', \mathcal{R}), \qquad \mathbf{D}' \in \mathbb{R}^{T\times m}$$

Context $\mathcal{C}$ carries symbol, timeframe, price column, horizon set, cost parameters, target, and auxiliary sources. Identified by unique name, registered by decorator, parameterized by a config section keyed to that name, persisted through *one* result schema. The uniformity is the point: a new analysis is a registry entry, not a new script.

The planted-signal discipline

Every unit ships behind a mandatory planted (synthetic) conformance gate **written before the implementation**. Not a style preference — three estimator bugs in Stage 1 were caught only this way, and nothing else would have caught them, because on real data a subtly wrong estimator still returns plausible numbers.

Why this is a paper and not just infrastructure

It supplies the theory that says what the other papers' numbers *mean*: #2's five gates are a special case of a process contract, #7's eight independent axes are an N_{eff} claim, #5's whole argument for an exogenous signal is an N_{eff} argument, and #8's gate ladder is this discipline applied one level down, to data rather than signals.

One-pager for personal use. The sentence to remember: *the binding correlation is the stress-regime correlation.* Everything else here is machinery for measuring it honestly. Detail: `docs/specs/process_layer.md` (PROC-1..18).